

Founder Signal — Product Requirements Document

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A scheduled daily research pipeline discovers signals, aggregates them into problems, ranks opportunities and publishes the strongest ones.

1. Information Architecture

- Home — today's strongest opportunities.
- Explore — category/score/momentum/competition filters.
- Opportunity Detail — visual evidence and build recommendation.
- Trends — momentum and category movement.
- Categories — category feeds.
- Methodology — scoring and data explanation.

2. Opportunity Scoring

Demand 30% + Momentum 25% + Problem Severity 20% + Commercial Intent 15% + Competition Gap 10%. The score is calculated in code; AI explains it.

3. Daily Pipeline

Stage	Action	AI?	Storage
Ingest	Collect new items	No	Transient
Filter	Rules + metadata	No	Transient
Dedup	Hash + fuzzy	No	Transient
Classify	Intent heuristics	Optional	Transient
Aggregate	Counts + trends	No	Permanent
Pre-score	Rank candidates	No	Permanent
Enrich	Top opportunities	Yes	Permanent
Publish	Update website	No	Permanent

4. UI Requirements

- Visual-first opportunity cards.
- Score, demand, momentum and competition immediately visible.
- 3–4+ useful visualizations on detail.
- Plain-language explanations.
- Metric tooltips.
- Progressive disclosure.

5. Lightweight Architecture

- Python + FastAPI.
- PostgreSQL.
- Next.js + Tailwind.

- Scheduled Python workers.
- No vector DB required for MVP.
- No Kafka/Kubernetes/Airflow/data warehouse.
- Batch LLM enrichment only for top clusters.
- Raw source payloads use TTL.

6. Acceptance Criteria

- Daily publishing without user intervention.
- Opportunity understood within 30 seconds.
- Detail page creates trust without raw posts.
- Duplicate ingestion cannot materially inflate signal counts.
- LLM failure does not stop signal updates.
- Source failure is isolated.